```
.loop
    mov     dl,byte [rsi]
    xor     dl,byte [rdi]
    inc     rsi
    inc     rdi
    or      al,dl
    loop    .loop
```

rdx

dh dl

dx

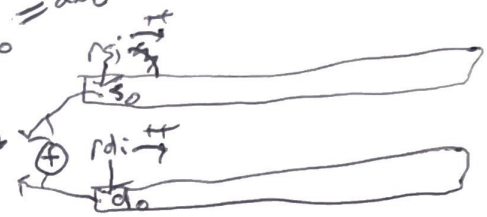
[0x11]

$dil \in rsi \longrightarrow dl \;\; \geq d_0 = dl_0$

$\{ dil \in rdi \;\; vs \;\; dl \}$ rdx $\;\; \| \;\; dl$

$\perp$

dl

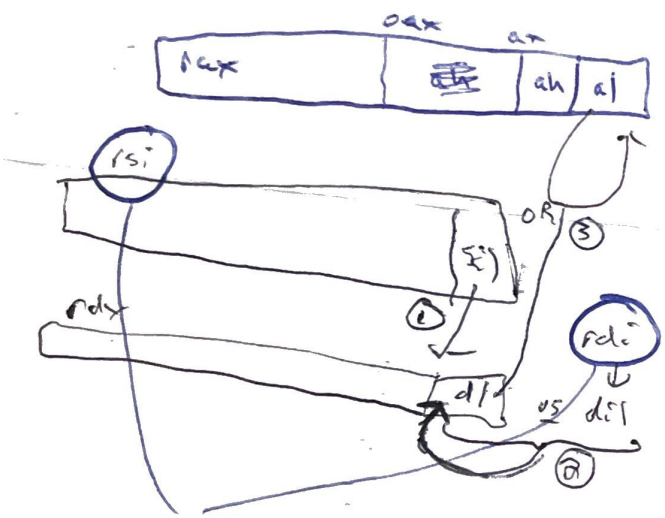


$= al_0 \vee dl_0 = al_0 \vee (s_0 \oplus d_0)$

$= al_1 \vee (s_1 \oplus d_1) = (al_0 \vee (s_0 \oplus d_0)) \vee (s_1 \oplus d_1)$

$= al_0 \vee \bigvee_{j=0}^{rcx-1} (s_j \oplus d_j) = 0 \Leftrightarrow [rsi+i] \| [rdi+i] \;\; \forall i \in [0, rcx]$

difference check



(rcx) .loop:

```
    mov     dl,byte [rsi]
    xor     dl,byte [rdi]
    inc     rsi
    inc     rdi
    or      al,dl
    loop    .loop
```

compares rsi & rdi

(stores result ∈ rax),

loop → byte-by-byte checking

[0x11]